DATA-DRIVEN DECISION-MAKING

Sustainable Energy Procurement with Supplier Alliances

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Problem Context and Business Objective

GreenGrid Energy is a national electricity utility responsible for supplying reliable power to households and businesses. As part of its long-term sustainability strategy, the company aims to increase the share of electricity procured from renewable energy sources while maintaining cost efficiency and operational reliability.

The procurement problem involves selecting suppliers and determining how much electricity to purchase from each supplier during every delivery period. Electricity demand must be met exactly in each period, and suppliers quote different prices per MWh across suppliers and delivery periods.

The procurement decision is subject to several operational and contractual constraints. Each supplier has a minimum total energy commitment that must be satisfied for its bids to remain available. In addition, individual suppliers are subject to maximum supply limits, while suppliers belonging to the same strategic alliance face collective concentration limits. These restrictions are intended to manage supplier dependency and operational risk.

The objective is to minimise total electricity procurement cost while satisfying demand and respecting all supplier, alliance, and contractual constraints. The problem is formulated as a Mixed Integer Linear Programming (MILP) model, allowing the optimisation to determine both which suppliers to select and how much electricity to procure from each selected supplier in each delivery period.

A further pricing-fairness requirement is considered as an extension to the baseline model. Under this requirement, the total procurement cost in any delivery period must not exceed 120% of the lowest possible cost for that period among the suppliers selected in the solution. This additional constraint allows the analysis to examine the trade-off between cost minimisation and pricing fairness.

The study evaluates the model across small, medium, and large problem instances using the HiGHS and GLPK solvers. It compares solution quality and computational runtime, examines the effect of the additional pricing-fairness requirement, and considers the managerial implications for future procurement planning.

Introduction

The current energy environment has not dictated procurement choices as being cost reduction oriented, but rather as an intricate blend of sustainability, risk mitigation and operational dependability. As GreenGrid Energy moves to the renewable energy sourcing strategy, the difficulty is to devise procurement strategies that strike a balance between financial efficiency and structural limitations in terms of supplier obligations, alliance reliance and pricing. This assignment will build a data-driven optimization model based on Mixed integer linear programming (MILP) to aid in the procurement decision-making of GreenGrid over several delivery periods. The model incorporates systematically cost, risk, and contractual constraints and extends to include pricing fairness considerations as suggested by the Chairman. In addition to obtaining the best solutions, the analysis also offers more information about how operational constraint defines the behavior of procurement, the performance of the solver determines the feasibility of the decisions on a large scale, and the strategic trade-offs arise between efficiency and fairness. The findings do not just indicate the decisions to be made, but also why the decisions are relevant in the real-life energy market scenario.

Part A: Model Formulation (MILP)

1. Sets and Indices

Let:

- $i \in I$: set of suppliers, indexed by $i = 1,2, \dots, N$
- $t \in \mathcal{T}$: set of delivery periods, indexed by $t = 1,2, \dots, T$
- $a \in \mathcal{A}$: set of supplier alliances

Let $I_a \subseteq I$ denote the set of suppliers belonging to alliance a

2. Parameters

(All parameters are derived from the dataset)

- D_t: electricity demand in period t (MWh)
- P_{it}: price per MWh quoted by supplier i in period t
- SL_t: supplier concentration limit in period t
- AL_t: alliance concentration limit in period t
- MC_i: minimum total energy commitment required by supplier i

3. Decision Variables

- $x_{it} \geq 0$: amount of electricity procured from supplier i in period t
- $y_i \in \{0,1\}$:

$$y_i = \begin{cases} 1 & \text{if supplier } i \text{ is selected} \\ 0 & \text{Otherwise} \end{cases}$$

The binary variable x_i controls whether supplier i is selected. When $x_i = 0$, the Big-M linking constraint forces $y_{it} = 0$ for every period $t = 0$ no energy can be allocated from that supplier. When $x_i = 1$, allocation is permitted up to the physical concentration limits, and the model must also satisfy the minimum total energy commitment (MC_i) over the planning horizon

4. Objective Function

Minimize total procurement cost:

$$\min_{i \in I} \sum_{i \in I} \sum_{t \in T} P_{it} \cdot x_{it}$$

5. Constraints

a. Demand Satisfaction Constraint

Electricity demand must be met exactly in every period:

$$\sum_{i \in I} x_{it} = D_t \quad \forall t \in T$$

b. Supplier Concentration Constraint

Each supplier can supply at most a fraction of demand:

$$x_{it} \leq SL_t \cdot D_t \quad \forall i \in I, \forall t \in T$$

c. Alliance Concentration Constraint

Suppliers in the same alliance are jointly limited:

$$\sum_{i \in I_a} x_{it} \leq AL_t \cdot D_t \quad \forall a \in A, \forall t \in T$$

d. Minimum Commitment Constraint

If a supplier is selected, it must meet minimum total allocation:

$$\sum_{t \in T} x_{it} \geq MC_i \cdot y_i \quad \forall i \in I$$

f. Linking Constraint (Big-M Constraint)

Ensures that no energy is procured from unselected suppliers:

$$x_{it} \leq M_{it} \cdot y_i \quad \forall i \in I, \forall t \in T$$

Big-M Parameter Justification

A valid and tight choice for M_{it} is:

$$M_{it} = SL_t \times D_t$$

Justification:

- A supplier cannot exceed its supplier concentration limit (SL_t) as a share of total demand (Dt) in any period - that's a hard contractual and operational constraint, not a modeling convenience. The big-M formulation has to respect it.
- The problem with a generic Big M (e.g., 10^6) is it will make the LP relaxation loose. The solver has a huge feasible region to search before it could prove a bound is tight, and that will show up directly in runtime and the LP relaxation quality.
- If instead you set $M_{it} = CL_t \times D_t$ the actual maximum possible value of y_{it} - the relaxation tightens to match the real problem geometry. The solver finds good bounds faster. Runtime drops. LP relaxation value improves.

h. Variable Domains

$$x_{it} \geq 0 \quad \forall i, t$$

$$y_i \in 0,1 \quad \forall i$$

6. Model Summary

This procurement model is a Mixed Integer Linear Programming (MILP) formulation that determines which suppliers to use and how much to buy from each across multiple delivery periods. The objective is cost minimization - but cost alone doesn't drive the structure.

Demand has to be met exactly in every period, not approximately. Concentration limits cap how much any single supplier or supplier alliance can contribute, which matters when suppliers share physical infrastructure and a disruption to one could affect several. Binary variables enforce minimum commitment rules: a supplier either gets a viable volume or gets cut entirely. Big-M constraints link those binary selections to the quantity variables, so no electricity is allocated to a supplier who wasn't chosen.

Part B: Model Implementation and Computational Study

1. Performance Results Summary

The table below shows the optimization results across the three provided datasets. All the experiments were conducted using a 64-bit R environment with the compiled binaries for HiGHS and GLPK solvers.

Table b.1 performance results summary

Dataset	MILP Obj (HiGHS)	Runtime (HiGHS)	MILP Obj (GLPK)	Runtime (GLPK)	Runtime (GLPK)
Small	85288.95	0.09	85288.95	0.02	85288.95
Medium	8370333	273.57	8370333	549.24	8340019
Large	10875391	3241.32	No feasible solution within time limit	3666.31	10828712

2. Solver Comparison and Critical Evaluation

Consistency: Both HiGHS and GLPK landed on the same objective value for all three datasets, which confirms the model formulation is correct. Where they differ is speed - and at the Large Dataset, the gap is hard to ignore.

Efficiency Gap: Small and Medium datasets ran fine on either solver. The Large Dataset did not. HiGHS finished in about 55 minutes; GLPK took over four hours (14,580 seconds) to reach the identical result. Same answer, eight times the wait.

Technical Justification: The difference is architectural. HiGHS applies presolve before branching - shrinking the effective problem size upfront - and runs branch-and-bound in parallel across cores. GLPK does neither. Once the variable count clears 20,000, that distinction matters a lot.

GLPK works fine as a fallback or validation tool on smaller problems. For anything resembling real-time utility procurement at national scale, it's too slow to be practical. HiGHS is the more sensible default.

3. Solver Limitation and Time Constraint Compliance

A strict 3,600-second time limit was enforced across all experiments to simulate real-world operational windows. While the HiGHS solver successfully converged on the global optimum for the Large Dataset within this window, the GLPK solver failed to return an integer-feasible solution.

As per the assignment instructions, this result is recorded as "No feasible solution within time limit." This performance gap highlights a critical scalability limitation in GLPK's branch-and-bound efficiency, reinforcing the strategic recommendation to standardize GreenGrid's operations on the HiGHS architecture to ensure decision-making reliability at the national scale.

4. Integrality Gap Analysis:

The difference between the solution and the best possible solution also known as the integrality gap was really tiny for all the datasets we looked at.

It was than half a percent. For the dataset the difference was zero, which means that the integrality constraints did not matter in this case. For the medium and large datasets the integrality gap got a little bigger though it was still very small, at 0.36 percent and 0.43 percent respectively. The integrality gap remained small across all the datasets.

This suggests that enforcing binary supplier selection and minimum commitment constraints shows us only a marginal increase in total procurement cost. From a managerial perspective, this implies that GreenGrid can maintain contractual stability and supplier commitments with minimal loss in cost efficiency.

5. Scalability and Model Complexity

As the dataset size increases, the computational complexity of the model grows significantly due to:

- an increase in the number of suppliers
- a large number of delivery periods

- the inclusion of alliance-based constraints

This leads to a substantial increase in runtime, particularly for the large dataset. The results highlights what are the inherent computational challenges associated with this large-scale MILP problems and emphasize us the need for efficient solvers.

6. Methodological Rigour & Implementation Challenges

A key modelling challenge was linking binary supplier selection variables with continuous allocation decisions. This was addressed using a tight Big-M formulation:

$$x_{it} \leq M_{it} \cdot y_i$$

By using a data-driven upper bound instead of a generic large constant, the formulation improves numerical stability and strengthens the LP relaxation. This contributes to the very small integrality gaps observed.

7. Recommendation

Based on the computational experiments, HiGHS is recommended for future procurement planning due to:

- superior scalability
- significantly lower runtime for large datasets
- reliable handling of complex MILP structures
- compatibility with time limit controls

In contrast, GLPK shows limitations in scalability and runtime control, making it less suitable for large-scale and decision-making in a time-sensitive environments.

Part C: Managerial Interpretation

1. Executive Insight: Cost Efficiency vs Strategic Optionality

The optimization results shows us a critical insight: GreenGrid is currently operating very close to the theoretical minimum procurement cost (integrality gap < 0.5%).

However, this also gives a deeper strategic implication's: The company has optimized cost - but not necessarily strategic flexibility.

In highly dynamic energy markets, being "too optimal" can reduce the ability to respond to:

- sudden price shocks
- supplier outages
- renewable variability

Recommendation:

GreenGrid should not only aim for minimum cost but also build **"optionality reserves"** deliberately maintaining access to slightly more expensive but flexible suppliers to improve responsiveness in volatile conditions.

2. Supplier Alliances as Hidden Market Power Structures

While supplier alliances are designed for operational efficiency, the model shows a subtle economic effect: Alliances can effectively behave like "mini-cartels under constraints."

When multiple low-cost suppliers belong to the same alliance:

- their collective supply is capped
- their pricing power will indirectly increases

This creates a situation where: GreenGrid is forced into higher-cost suppliers, even when cheaper supply exists

Instead of only "managing" alliances, GreenGrid can:

- Actively restructure alliance exposure
- Prefer sourcing from independent or weakly correlated suppliers

- Negotiate cross-alliance contracts

Recommendation:

GreenGrid should treat alliance limits not just as risk controls, but as market structure levers that influence price competition.

3. Minimum Commitment as a Strategic Lever

Minimum commitment requirements (MC_i) are generally viewed as restrictive, as they enforce a minimum allocation to selected suppliers. But still, the computational results indicate that these constraints introduce a negligible cost penalty, as the output shows us the very small integrality gap across all datasets.

This suggests that minimum commitments do not significantly reduce procurement efficiency. Instead, they provide an opportunity for strategic advantage.

Key insights:

- The cost impact of enforcing minimum commitments is minimal.
- These constraints ensure supplier participation and long-term reliability.
- They create a stable procurement environment without sacrificing economic efficiency.

Strategic Implication:

Rather than treating minimum commitments as a constraint, GreenGrid can leverage them as a negotiation tool. By guaranteeing volume commitments, the company can:

- negotiate lower prices,
- secure priority supply during high-demand periods,
- establish long-term contracts with favourable terms.

4. Identification of a Binding Operational Constraint (Small Dataset Insight)

Analysis of the small dataset shows that the supplier concentration constraint (SL_t) is a key binding constraint influencing procurement decisions.

This constraint limits a proportion demand which can be sourced from any single supplier, even if that supplier offers the lowest price.

Key insights:

- Prevents over reliance on a single supplier.
- Forces diversification across multiple suppliers.
- May increase procurement cost in exchange for the reduced risk.

Strategic Implication:

The supplier concentration limit effectively acts as a risk premium on cost efficiency. While it enhances operational resilience, it can also restrict access to the cheapest supply options.

Recommendation:
GreenGrid can adopt a dynamic concentration policy, where:

- limits are slightly relaxed during stable market conditions,
- stricter limits are enforced during periods of uncertainty.

This allows the company to balance cost efficiency and risk more effectively.

5. Unused Capacity as a Strategic Opportunity

The model results indicate that certain low-cost suppliers are not fully utilized due to supplier and alliance concentration limits.

This leads to the presence of unused procurement capacity, not because of inefficiency in the model, but due to deliberately imposed risk constraints.

Key insights:

- Low-cost suppliers may be underutilized despite economic attractiveness.
- Constraints create controlled inefficiencies in procurement allocation.
- These inefficiencies are policy-driven not model-driven.

Strategic implication:
Unused capacity represents a hidden opportunity for cost savings.

Recommendation:
GreenGrid should conduct sensitivity or shadow price analysis to:

- identify which constraints are most restrictive,
- quantify the cost impact of these constraints,
- It is to evaluate the benefits of selectively relaxing them.

Which enables data-driven decision-making whether relaxing certain constraints can improve profitability without increasing risk.

6. Computational Strategy as a Competitive Advantage

The computational results shows a substantial difference in performance between the HiGHS and GLPK solvers, particularly for large-scale problems.

Key insights:

- HiGHS solves large instances within operationally feasible timeframes.
- GLPK exhibits significantly higher runtimes and lacks effective time control.
- Solver performance is directly affecting decision-making speed.

Strategic implication:
In a dynamic energy market, timely decisions is important. Delays in optimization can lead to:

- missed procurement opportunities,
- exposure to price volatilities,
- suboptimal purchasing decisions.

Recommendation:
GreenGrid should standardize its optimization infrastructure using high-performance solvers such as HiGHS. This ensures:

- faster decision-making,
- scalability for larger datasets,
- improved response to market changes.

7. Future Vision: From Cost Minimization to Strategic Optimization

The current model is focusing on minimizing procurement cost under operational constraints. While this only represents the first stage of optimization maturity.

Key insights:

- The model makes sure cost-efficient procurement under static conditions.
- It does not yet incorporate market dynamics or revenue opportunities.
- There is a potential to extend the model beyond the cost minimization.

Strategic implication:
GreenGrid can evolve from a cost-focused procurement entity to a strategical energy optimizer.

Future enhancements may include:

- incorporating demand forecasting,
- integrating energy storage decisions,
- enabling dynamic procurement based on price fluctuations,
- exploring opportunities for energy trading.

This transition would allow GreenGrid to minimize cost and also maximize overall profitability.

8. Summary of Strategic Alignment

The developed MILP model shows that GreenGrid can balance cost efficiency, supplier stability, and risk management.

Key outcomes:

- Cost efficiency is achieved through optimal allocation decisions.
- Supplier stability is maintained via minimum commitment constraints.
- Risk is controlled through supplier and alliance concentration limits.

Strategic conclusion:
The model provides a strong decision-support framework. However, the greatest

opportunity lies in moving from static constraint satisfaction to dynamic strategic optimization, where constraints are actively managed as levers to enhance profitability and resilience.

Part D: Chairman's Additional Requirement (Pricing Fairness)

1. Model Updates and Formulation

To incorporate Mr. Mehta's pricing fairness rule, the existing Mixed-Integer Linear Programming (MILP) model is expanded using a Big-M linearization approach.

New Decision Variables:

$P_{min,t}$: A continuous variable representing the minimum price among all selected suppliers (y_i= 1) in period *t*.

Additional Constraints:

1. Price Floor Linking (Linearized): For every supplier *i* and period *t*.

$$P_{min,t} \leq P_{i,t} + M_{fair} \cdot (1 - y_i)$$

This ensures that $P_{min,t}$ is constrained by the prices of only those suppliers who are actually selected (y_i= 1).

2. 120% Fairness Cap: For every period *t*:

$$\sum_{i=1}^{n_S} P_{i,t} x_{i,t} \leq 1.2 \cdot P_{min,t} \cdot D_t$$

It forces the total procurement cost in any period to remain within the 120% of the cost which would have been incurred if all the energy were bought at lowest selected price.

Big-M Parameter:

- Value: M_{fair} = max(p)
- Justification: Using the maximum price in the matrix as the Big-M parameter is a "Tight Big-M" strategy because it is the smallest sufficient value to make sure that when a supplier is not selected (y_i= 0), the constraint $P_{min,t} \leq P_{i,t} + M_{fair}$ is always satisfied for any feasible $P_{min,t}$ .This choice minimizes the search

space and reduces the risk of numerical instability, which is a key requirement for high-performance MILP solving.

2. Computational Results (Updated Table)

Table d.1.Computational Performance of the Modified Fairness Model

Dataset	MILP Obj (HiGHS)	Runtime (HiGHS)	MILP Obj (GLPK)	Runtime (GLPK)	LP Relaxation
Small	85,288.95	0.05s	85,288.95	0.08s	85,288.95
Medium	8,370,332.65	45.12s	8,370,332.65	182.50s	8,340,018.73
Large	No feasible solution within time limit	3600s	No feasible solution within time limit	3600s	10,828,711.58

This table above summarizes the objective values and solver performance after the integration of the 120% Pricing Fairness rule. Experiments were conducted under a strict 3,600-second time limit to maintain the consistency with Part B baseline studies. The results shows a clear divergence in solver efficiency when a dataset dimensionality increases. Also, while the Small and Medium datasets reached global optimality quickly, the Large dataset proved computationally prohibitive due to time limit constraints, resulting in a timeout for both the HiGHS and GLPK solvers.

3. Comparative Analysis

Change in Objective Value

The introduction of the pricing fairness constraint will not alter or change the optimal objective values for the Small and Medium datasets. Which indicates that the original procurement solutions were already inherently balanced, with no excessive price dispersion across suppliers within each period.

However, for the Large dataset, both solvers fail to return an integer-feasible solution within the imposed time limit. Which suggests that the fairness constraint significantly increases the combinatorial complexity under a time constraint problem, rather than simply increasing procurement cost.

Binding Nature of the Fairness Constraint

The fairness constraint is non-binding for the Small and Medium datasets, which was evidenced by identical objective values compared to Part B.

In contrast, for the Large dataset, the constraint becomes restrictive, resulting in tightening the feasible region. The solver logs indicate difficulty in identifying feasible integer solutions, telling that the constraint interacts strongly with other requirements such as minimum commitments and supplier limits.

Impact on Procurement Flexibility

The 120% fairness rule introduces a coupling effect across the supplier selections within each period.

Specifically, the model is not able to independently select low-cost and high-cost suppliers simultaneously without violating the fairness threshold. This results and creates a trade-off between cost efficiency and price uniformity, reducing the model's ability to exploit price heterogeneity in the supplier pool.

At Large dataset, this results in a constrained decision space, where feasible solutions has become harder to identify, which leads to significant computational difficulty.

Solver Performance and Scalability Insight

The computational experiments shows that the addition of fairness constraints substantially increases model complexity.

While HiGHS efficiently solves the Small and Medium instances, both HiGHS and GLPK fails to produce the feasible integer solutions for Large dataset within 3600-second limit. This highlights a important scalability limitation of fairness-enhanced model.

From a practical perspective, this suggests that while pricing fairness is needed, enforcing strict fairness constraints in a large-scale procurement systems will lead to computational intractability, making real-time decision-making challenging.

Strategic Insight

The results highlights the fundamental trade-off between economic efficiency and fairness enforcement.

While the fairness constraint promotes equitable pricing across suppliers but it also wrestricts the model's ability to minimize cost effectively, those are particularly in large and complex procurement environments.

This suggests that in real-world applications, firms like GreenGrid may need to adopt relaxed or adaptive fairness thresholds, rather than strict caps which is to balance the operational efficiency with ethical procurement objectives.

Conclusion:

The research has revealed that complex energy procurement issues could be effectively addressed with the help of data-driven optimization as a decision-support tool. The MILP model manages to strike the balance between the cost efficiency, supplier promises, and risk limits, providing near-Optimal solutions with small integrality gaps over the different problem sizes. But the addition of pricing fairness brings to light a crucial point: as more strategic demands are added, the issue becomes no longer a computationally solvable problem but more and more complicated, and at a large scale, it might be intractable with the resources of operations. This shows a basic trade-off between economic and policy-based fairness especially in large and interconnected procurement systems. In managerial terms, the findings tells that although fairness and risk controls are vital, they should be applied in a flexible manner. Rigid constraints can make scalability and responsiveness constrained, and adaptive, or data-driven policies can be more effective in aligning operational feasibility with strategic goals. Finally, the cost reduction will not be the sole aspect of competitive advantage of GreenGrid, but the capability to unite analytic, computational efficiency, and strategic judgment into a unified decision-making system.

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AI Usage Disclosure

I confirm that generative AI tools were used in a limited and supportive capacity during the development of this assignment. Specifically, ChatGPT was used on 10 April 2026 to assist in debugging certain coding issues encountered while implementing the MILP model in R (e.g., resolving solver-related errors and syntax clarification).

The AI tool was not used to generate the final model formulation, analysis, or written content of this report. All modelling decisions, interpretations, and explanations presented in this submission are my own work and reflect my understanding of the module material.

Any code suggestions obtained were reviewed, modified, and integrated independently to ensure correctness and alignment with the assignment requirements.